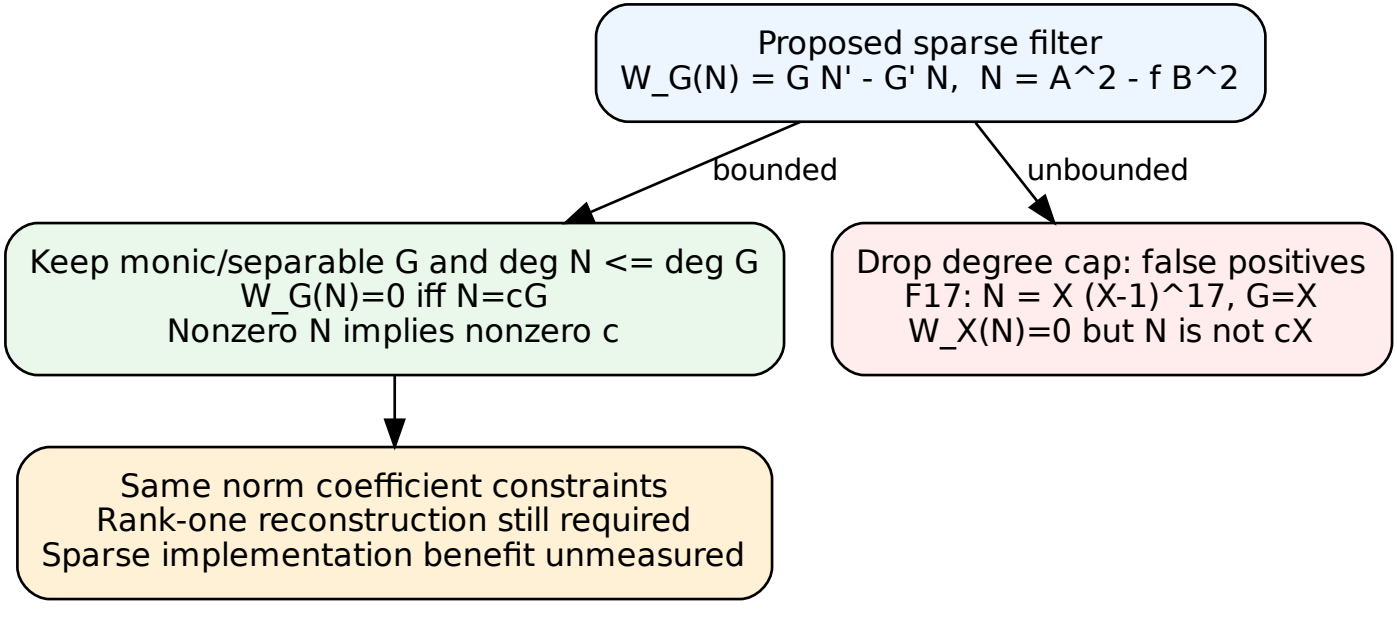
Proposed sparse filter
W_G(N) = G N' - G' N, N = A^2 - f B^2
bounded
unbounded
Keep monic/separable G and deg N <= deg G
W_G(N)=0 iff N=cG
Nonzero N implies nonzero c
Drop degree cap: false positives
F17: N = X (X-1)^17, G=X
W_X(N)=0 but N is not cX
Same norm coefficient constraints
Rank-one reconstruction still required
Sparse implementation benefit unmeasured